

EXTENSIONS OF THE TURING MACHINE

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LEARNING OBJECTIVE

- To Design Turing machines for any Languages (K3)
 - To Understand the concept of Turing Machine

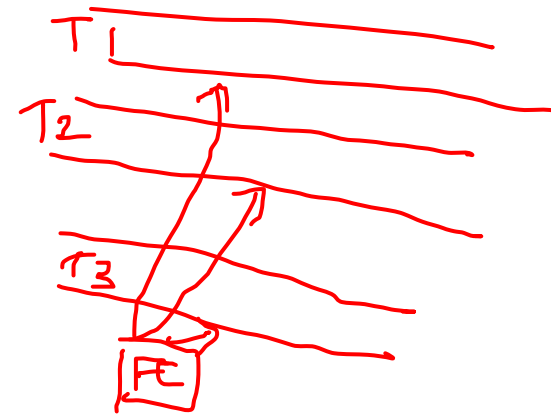
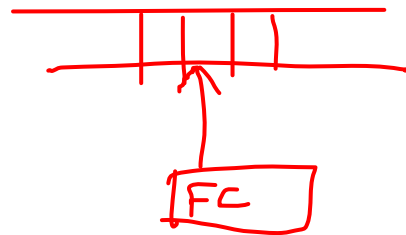
POSSIBLE EXTENSIONS

- Multiple tapes
- Two-way infinite tapes
- Two-dimensional tapes
- Multiple heads
- Random access
- Nondeterministic

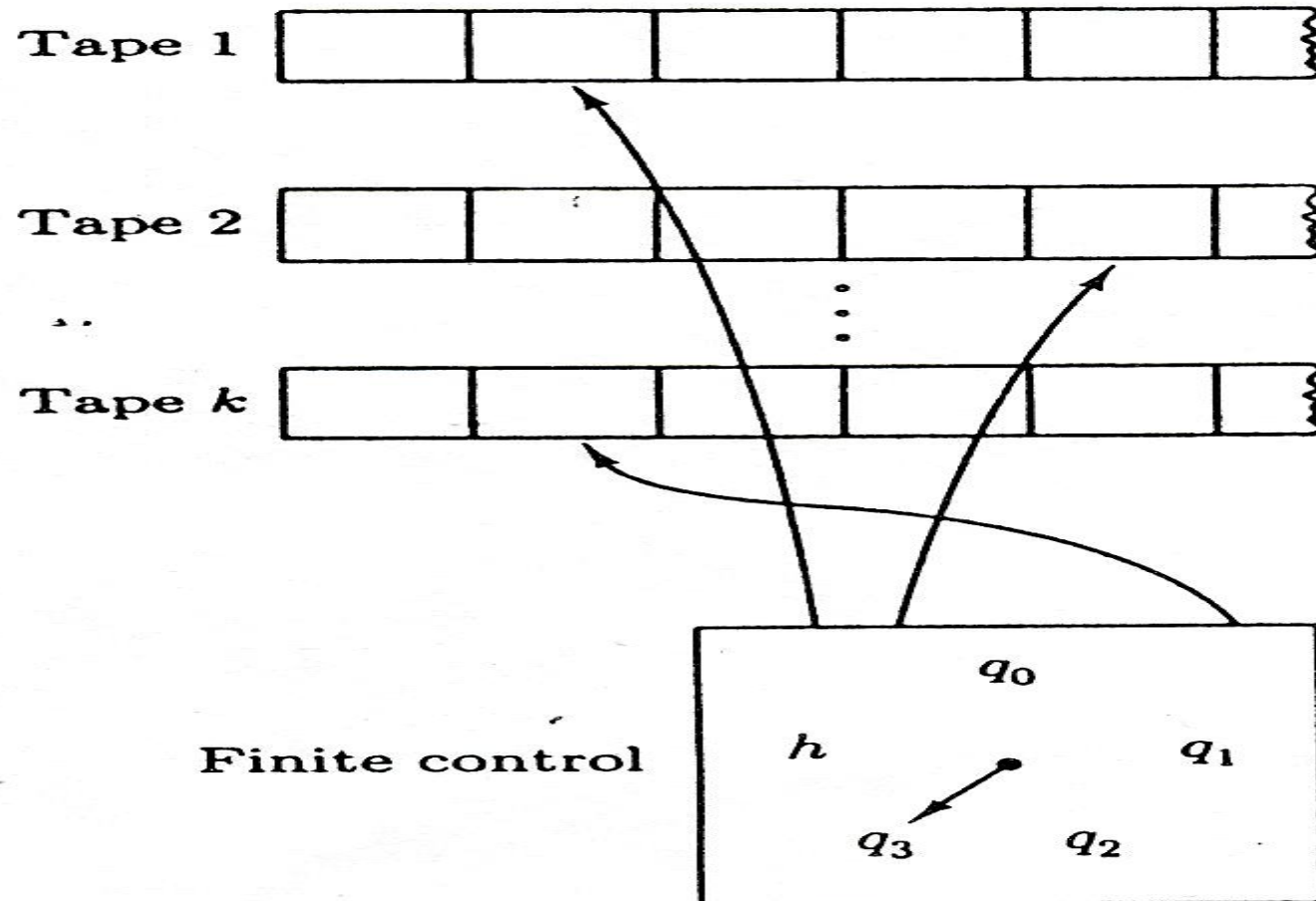
MULTIPLE TAPE TURING MACHINE

- Each tape is connected to the finite control by means of a read/write head
- For any fixed integer $k \geq 1$
 - A k-tape Turing machine is a Turing machine equipped with k tapes and corresponding heads

Multitrack
one head



MULTIPLE TAPE TURING MACHINE



EX FOR THE MULTIPLE TAPE TM

$x + y$

- x on first tape, y on second tape and results written to third tape
 - move 1 and 2 heads to right end, move head 3 to right $\max(|x|, |y|)$
 - move tape 3 head right one bit for overflow

x unary
↑

 y

result

EX FOR THE MULTIPLE TAPE TM

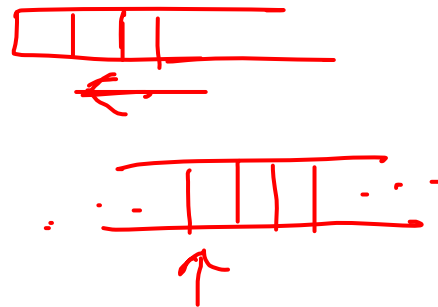
- add 1 and 2, bit by bit and writes each intermediate result as follows
 - reads bits at 1 and 2 plus carry from previous bits
 - if sum is 0 or 1, write it to tape 3
 - if sum is 2 or 3, set carry and write 0 or 1 on tape 3
- if one string ends (beginning of tape marker) use 0 for that input and do not move that head

USAGE OF MULTIPLE TAPE TM

- The use of a k-tape Turing machine:
 - computing a function
 - deciding or semideciding a language

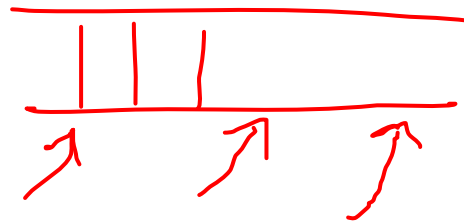
TWO-WAY INFINITE TM

- The tape is infinite in both directions
- All squares are blank (exception: those containing the input)
- It can be simulated by a 2-tape machine:
 - Tape 1: contains the part of the tape to the right of the square containing the first input symbol
 - Tape 2: contains the remaining part of the tape to the left.



MULTIPLE HEADS TURING MACHINE

- Uses a single tape and multiple heads
- In any state only one head can write or move
- The heads all sense the scanned symbols and move or write independently



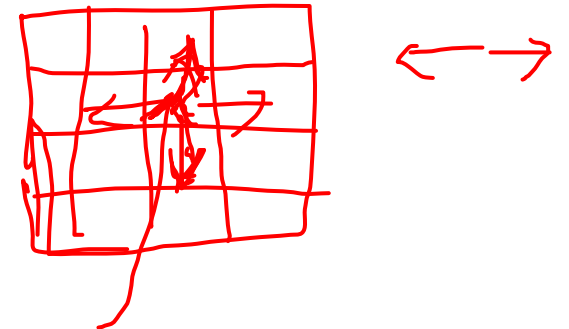
MULTIPLE HEADS TURING MACHINE

- $L = (a^n b^n c^n \mid n = 0, 1, 2, \dots)$
 - given string w , position first head at beginning of input
 - position second head past all a 's to the first b
 - position third head past all a 's and all b 's to first c
 - enter loop verifying that, on each iteration, head 1 reads an a , head 2 reads a b and head 3 reads a c
 - if third head reaches end of input string at the same time head 1 reads the first b and head 2 reads the first c , machine erases the input string and writes a 1 into cell 1 to signify acceptance

$a^2 b^2 c^2 \rightarrow a a b b c c$

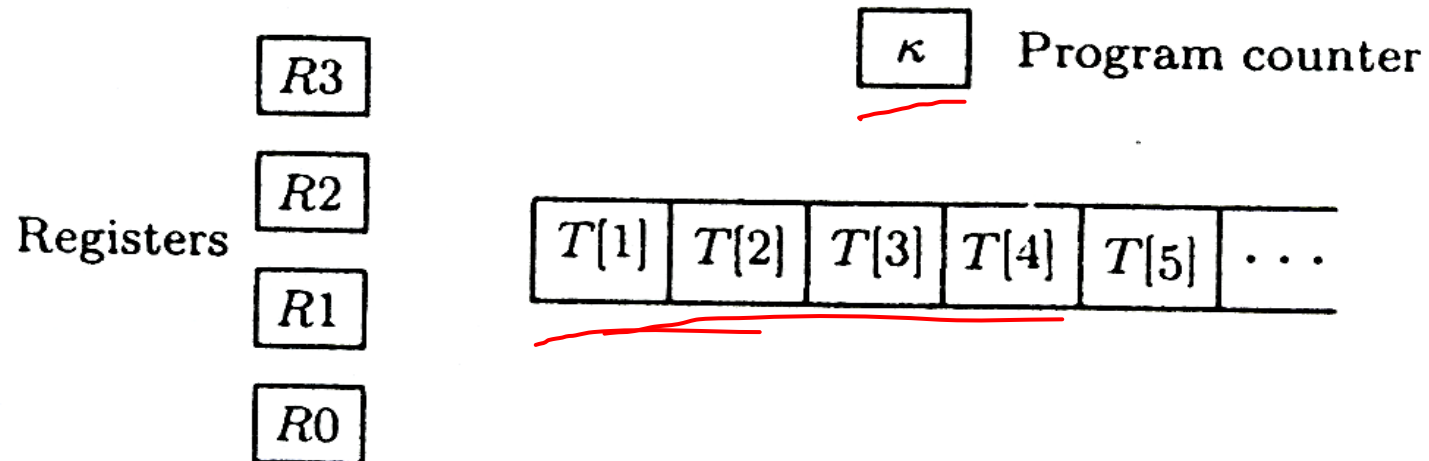
2D TAPE TURING MACHINE

- the input string is placed on the first tape, such as in case of a standard Turing machine
- tape: an infinite two-dimensional grid
- one head on a two dimensional grid that could expand indefinitely down and to the right
- head can move in four different directions
- end of tape markers on left and top sides



RANDOM ACCESS TURING MACHINES

- A random access Turing machine has:
 - a fixed number of registers
 - a one-way infinite tape
 - a program counter



SEQUENCE OF INSTRUCTIONS

Instruction	Operand	Semantics
read	<u>j</u>	<u>$R_0 := T[R_j]$</u>
write	j	<u>$T[R_j] := R_0$</u>
store	j	<u>$R_j := R_0$</u>
load	j	$R_0 := R_j$
load	=c	$R_0 := c$
add	j	$R_0 := R_0 + R_j$
add	=c	$R_0 := R_0 + c$
sub	j	$R_0 := \max \{R_0 - R_j, 0\}$
sub	=c	$R_0 := \max \{R_0 - c, 0\}$
half		$R_0 := [R_0 / 2]$
<u>jump</u>	s	<u>$k := s$</u>
<u>jpos</u>	s	if <u>$R_0 > 0$</u> then <u>$k := s$</u>
<u>jzero</u>	s	if <u>$R_0 = 0$</u> then <u>$k := s$</u>
halt		$k := 0$

SEQUENCE OF INSTRUCTIONS

- j stands for a register number, $0 \leq j < k$
- $T[i]$ denotes the current contents of tape square i
- R_j denotes the current contents of Register j
- $s \leq p$ denotes any instruction number in the program
- c is any natural number
- All instructions change k to $k+1$, unless explicitly stated otherwise

EXAMPLE

- program of a random access Turing machine, deciding the language $\{a^n b^n c^n : n \geq 0\}$.

acount := bcount := ccount := 0, n := 1

while T[n] = 1 do : n := n + 1, acount := acount + 1

while T[n] = 2 do : n := n + 1, bcount := bcount + 1

while T[n] = 3 do : n := n + 1, ccount := ccount + 1

if acount = bcount = ccount and T[n] = 0 then accept else reject

EXAMPLE

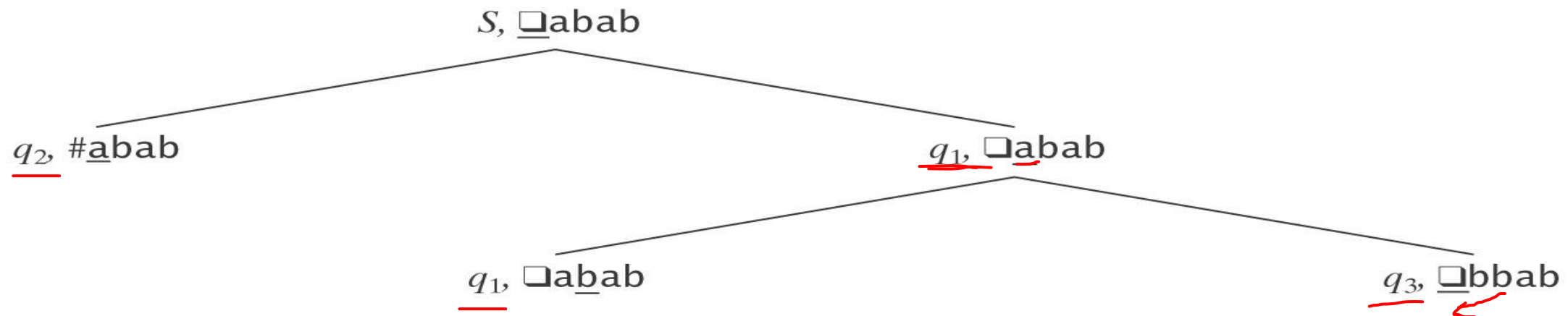
- We are assuming here that $E(a) = 1$, $E(b) = 2$, $E(c) = 3$
- We are using the variables `acount`, `bcount`, and `ccount` to stand for the number of a's, b's, and c's
- We are also using the abbreviation `accept` for "load =1, halt" and `reject` for "load =0, halt"

NONDETERMINISTIC TURING MACHINES

- At any state it is in and for the tape symbol it is reading, can take any action selecting from a set of specified actions rather than taking one definite predetermined action.
- Formally a **nondeterministic Turing machine** is a Turing machine whose transition function takes values that are $Q \times \Gamma \rightarrow \text{subsets of } (Q \times \Gamma \times \{L, R\})$

$$\begin{array}{c} Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\} \\ \uparrow = \\ 2 \end{array}$$

NONDETERMINISTIC TURING MACHINES



SUMMARY

- Extensions of Turing Machine

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008